

Le système international d'unités The International System of Units

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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 1

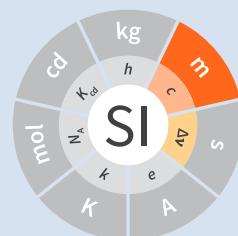
Part 1

Iodine ()

Iodine ()

CCL-CCTF Frequency Standards Working Group

May 2016
mai 2016



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 1: Iodine ()**

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1. Recommended value [1]

$$f(^{127}\text{I}_2) = 564\,074\,632.42 \text{ MHz}$$

equivalent to

$$\lambda = 531\,476\,582.65 \text{ fm}$$

with an estimated relative standard uncertainty of 1×10^{-10} applies to the radiation of a frequency-doubled diode DFB laser, stabilized with an iodine cell external to the laser with the following parameters:

- cold-finger temperature $(25 \pm 0.5)^\circ\text{C}$ (corresponding to the iodine pressure $p = 41 \text{ Pa}$)
- frequency modulation width, peak-to-peak, $(12 \pm 1) \text{ MHz}$ for $3f$ detection cases;
- saturating beam intensity of 12.7 mW ($\pm 10\%$) at a beam diameter of 1 mm

2. Source data

Adopted value $f = 564\,074\,632.42\text{ MHz}$
 $u_c/y = 1 \times 10^{-10}$

taken from:

f/kHz	u_c/y	source data
564 074 632 419 (8)	1.4×10^{-11}	[2]

This value was issued from only a single laboratory under conditions where strong linear absorption at Doppler centre occurs, thereby degrading the signal-to-noise level at this high iodine vapour pressure [2]. Thus, the CIPM following a recommendation of the CCL considered it prudent to enlarge the standard uncertainty by a factor of seven, and round to 1×10^{-10} .

Absorbing atom $^{127}\text{I}_2$, a_1 component, R(36) 32-0 transition

References

- [1] CIPM Recommendation 2 (CI-2015): Updates to the list of standard frequencies
<http://www.bipm.org/jsp/en/CIPMRecommendations.jsp>
- [2] T. Kobayashi, D. Akamatsu, K. Hosaka, H. Inaba, S. Okubo, T. Tanabe, M. Yasuda, A. Onae, F.-L. Hong, Compact iodine-stabilized laser operating at 531 nm with stability at the 10^{-12} level and using a coin-sized laser module, *Optics Express*, **23**, 20749 (2015).